Digital_to_CDP (CLMNBA-7870)

[CLMNBA-5148] [FR] Digital to CDP | IAM - Data - Login Interactions (Server-Side)

Created: 07/Jan/26 Updated: 02/Oct/26

Status:	READY FOR CONFIRMATION
Project:	Ignite (CMID)
Components:	CDP, R3.1_Business_Requirements
Affects versions:	None
Fix versions:	None
Parent:	Digital_to_CDP

Type:	Story	**Priority:**	Minor
Reporter:	Mushfiq Abdurrahman	**Assignee:**	John Mathew
Resolution:	Unresolved	**Votes:**	0
Labels:	CIAM, Digital_to_CDP, Ignite_Design_3.1		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Attachments:	image-20260424-025756.png image-20260424-025857.png
Issue links:	**Relates** relates to CLMNBA-5932 [FR] Digital to CDP | IAM - Configura... READY FOR CONFIRMATION

Acceptance Criteria:

Feature: Authenticated Login Identity Ingestion and Resolution

Scenario 1 — Identity mapping on successful authentication

Given A user authenticates successfully with Auth0

When the authentication event indicates **Login – Success**, **Signup – Success**, or **Silent Authentication – Success**
Then a Segment `identify()` and `track()` calls must be sent to Segment CDP
And the Auth0-generated `user_id` must be mapped to `userId`
And the login email address (username) must be mapped to `traits.email`
And identity values must be normalised and persisted on the Segment customer profile

Scenario 2— Session and profile continuity

Given A user has anonymous activity before logging in

When A successful authentication triggers a Segment `identify()` call

Then Anonymous and authenticated activity must be linked to the same profile
And the user must resolve to a single, persistent master customer ID (for example `subject_id`)

Scenario 3 — Tracking non-identity authentication events

Given An authentication-related event does not confirm a known user identity

When The event is one of the following:

- Login – Failure
- Login Setup – Failure
- Logout – Success or Failure
- Silent Authentication – Failure
- Token Exchange – Success or Failure
- Login – Notification / Warning
- Action Execution – Failure

Then A **Segment** `track()` **call** must be sent
And the event must not create or update customer identity
And the event must only be linked to a profile if a `userId` is already known
And the event must resolve to the existing master customer ID where available

Scenario 4 — Schema validation using Segment Tracking Plans

Given Auth0 events are ingested into Segment

When An event is received by Segment

Then The event must be checked against the **Segment Tracking Plan (Protocols)**
And required fields, data types, and formats must be validated
And any schema issues must create a **Protocols violation**

Scenario 6 — Real-time data delivery

Given An Auth0 authentication event occurs

	When The event is sent to Segment **Then** The event must be available in Segment **in real time or near real time** **And** identity resolution and downstream tools must receive timely updates **Scenario 7 — Monitoring and alerting** **Given** Auth0 events are flowing through Segment **When** There are issues such as: • Event volume drops or spikes • Delivery failures • Schema violations **Then** Segment Alerts must notify the relevant support teams **And** an incident must be automatically created in **BMC Helix** for investigation and resolution
Epic Link:	Digital_to_CDP
System:	Auth0

Description

HDBRD: CDP: 29.0, 30.0, **BRS:** BR060, BR061, **Brands:** Vodafone, **Component:** Auth0, **MoSCoW:** Must Have (9)

Description: *To enable unified identity resolution, contextual decisioning and behavioural insights, customer identifiers and interaction events captured during the login process via Identity Management platform will be delivered, normalised, and mapped to persistent master customer IDs within Segment CDP for Vodafone brand.*

Baseline Requirements:

User Story Format:

As a MarTech PO,
I want to capture and persist customer login events, user identifiers, and associated interactions, and match them against identified individual customers,
So that known customers can be uniquely resolved to a persistent master ID, enabling consistent identity resolution across channels, accurate reporting of login behaviour, contextual decisioning, and insights.

The Solution Must Format:

Integration Methods:

- The solution must support one of the below integration methods:
  - **Option A:** Native Auth0 → Segment (Cloud Destination) + Segment Destination Insert Functions
    - **Auth0 emits login events** (identify/track) to Segment using the native Auth0-to-Segment connection.
    - **Auth0 Rules/Actions/Hooks enrich the payload at source** with required identifiers (e.g., `user_id`, `email`) before delivery to Segment.
    - **Segment Destination Insert Functions perform downstream-specific transformation at egress**, including normalisation, schema alignment, enrichment, and conditional routing before forwarding events to each destination
    - This approach centralises “last-mile” logic inside Segment CDP, creating reusable transformation patterns across destinations and improving consistency of outbound payloads.
-
  - **Option B:** CIAM Integration via AWS Lambda → Segment HTTP Tracking API
    - **Auth0 post-login hooks invoke an AWS Lambda** function.
    - **Lambda transforms and normalises login metadata** (schema mapping, identity field selection, enrichment lookup if required).
    - Lambda forwards standardised events to Segment using the **HTTP Tracking API** (identify/track).
  - The solution must extend and apply these events to the **Vodafone brand** configured within the **Segment Vodafone workspace**.

Identity Mapping

- The solution must issue a **Segment** `identify()` **and** `track()` **calls** on successful authentication events (Login, Signup, and Silent Authentication) to establish or reinforce a known customer identity:
  - **Login – Success** (**Login Completed**)
  - **Login Setup – Success** (**Login Completed**)
  - **Silent Authentication – Success** (Session Refreshed)
- During these events, the solution must capture and persist the following identity attributes:
- `user_id`
  - Auth0-generated immutable identifier (for example: `auth0|66c82e7b6231a2d8cb0f3468`)
  - Mapped to the `userId` field in the Segment `identify()` call
  - Used as the **primary and persistent customer identity key**
- `email`
  - Login email address available at authentication time (for example: `john@domain.com`)
  - Mapped to the reserved `traits.email` field in the Segment profile
  - Treated as a mutable contact attribute, not a primary identifier

These attributes must be:

- Normalised (consistent naming, casing, and format)
- Persisted on the Segment profile
- Used to consistently resolve authenticated users to a unified customer identity across channels

The solution must also:

- Support **aliasing of anonymous and authenticated sessions** to enable session-to-profile continuity
- Resolve captured identifiers to a **persistent master customer ID** (e.g. user_name, subject_id)

Interaction Events

The solution must issue a **Segment** `track()` **call** for authentication-related interaction events that **do not establish a confirmed user identity** but are required for behaviour analysis and operational insights.The following events must be captured as **interaction events**, including but not limited to:

- **Login – Failure** (Login Failed)
- **Login Setup – Failure** (Login Setup Failed)
- **Logout – Success** (Logout Completed)
- **Logout – Failure** (Logout Failed)
- **Silent Authentication – Failure** (Session Refresh Failed)

These events must:

- Be recorded using **Segment** `track()` **calls**
- Include relevant context and error metadata where available
- Be associated to an existing user profile **only if a** `userId` **is already known**
- Not trigger identity creation or modification
- Resolve to to a **persistent master customer ID** (e.g. user_name, subject_id)

Schema Validation, Data Delivery & Alerts

- The solution must ingest Auth0 authentication events into Segment **in real time or near real time**, so identity resolution and downstream tools stay up to date.
- The solution must use **Segment Protocols (Tracking Plans)** to check Auth0 events before they are used for identity resolution or sent to destinations. The Tracking Plan must validate Auth0 event payloads for:
  - **Required fields**
  - **Correct data types**
  - **Allowed formats / values**
- When an Auth0 event does not match the Tracking Plan, it must create a **Protocols violation** (and can be controlled using Protocols controls/transformations as configured).
- The solution must use **Segment Delivery Overview** to provide visibility from **ingestion to destination delivery**, including:
  - events received
  - events filtered at source (including Protocols/Tracking Plan filtering)
  - events filtered at destination (including filters and Insert Functions)
  - failed deliveries and **events pending retry**
- The solution must configure **Segment Alerts** (default and/or custom) to notify relevant teams when IAM (Auth0) interaction events drops, including:
  - **Auth0 source volume** drops/spikes
  - **destination successful delivery rate** drops
  - **Protocols violations detected** (schema issues)
- Alerts must be routed to the relevant support teams and automatically create an incident in **BMC Helix** for investigation and resolution.

Solution Assumptions:

Integration Methods

- Auth0 login events will be delivered to Segment CDP using **Segment-supported ingestion patterns** (native cloud integration or HTTP Tracking API), emitting standard **Segment Spec** event types (`identify`, `track`, `alias`). [twilio.com]|https://www.twilio.com/docs/segment/connections/spec/identify]
- For **Option A**, identity enrichment performed in Auth0 (via Rules / Actions / Hooks) is assumed to be **lightweight and deterministic**, providing only required identifiers (e.g. `user_id`, `email`) prior to ingestion into Segment.
  - Destination-specific shaping, enrichment, or compliance logic will be handled **within Segment using Destination Insert Functions at egress**, not in the source system. [[twilio.com]|https://www.twilio.com/docs/segment/connections/functions/insert-functions]
  - Destination Insert Functions are assumed to execute **in Segment's cloud environment, after ingestion and identity resolution, and immediately before delivery to outbound destinations**, providing a reusable "last-mile" transformation layer across tools.
- For **Option B**, AWS Lambda is assumed to act as the **primary transformation and normalisation layer** prior to Segment ingestion, with events forwarded to Segment using the **HTTP Tracking API** and conforming to the Segment Spec.

Identity Resolution & Profile Management

- `user_id` (Auth0-generated identifier) will be treated as a **join key** within Segment and supplied as the `userId` field on `identify` events, consistent with Segment identity best practices.
- `user_name` (login email address) will be mapped to `email` in Segment profiles, enabling consistent downstream usability. [[twilio.com]|https://www.twilio.com/docs/segment/connections/spec/identify]
- Segment Unify will be the **system of record for identity stitching**, resolving anonymous and authenticated identifiers according to configured merge rules.

- Session-to-profile continuity will be achieved by supporting **aliasing and/or implicit merging** of anonymous activity into authenticated profiles when `identify()` is called post-login.
  Explicit `alias()` calls may be used where required for specific source patterns or downstream compatibility.
  [[twilio.com]|https://www.twilio.com/docs/segment/connections/spec/alias]
- A persistent master customer identifier (e.g. `email_name`) is assumed to be **stored as a profile trait** within Segment once resolved or enriched, and treated as an output of identity resolution rather than a required input at login.

Schema Validation, Data Delivery & Alerts

- A **Segment Tracking Plan (Protocols)** will be applied to the Auth0 source to validate login and identity events, including required identifiers, traits, and data types. [[twilio.com]|https://www.twilio.com/docs/segment/protocols]
- **Segment Protocols and Source Alerts** will be used to detect data-quality and ingestion issues such as missing identifiers, schema violations, and abnormal Auth0 event volumes. [twilio.com]
- Alerts will be sent via **email to BMC Helix**, enabling incidents to be raised and investigated promptly without the need for custom monitoring solutions.

Out of Scope:

-

Operational Benefits:

- Enables real-time identity resolution at login using deterministic identifiers.
- Supports unified customer profiles by linking online activity to offline master IDs.
- Reduces identity fragmentation across sessions, devices, and platforms.
- Ensures consistent login event data across Segment, Splunk, and SIEM

Supporting Documentation / Vendor Notes:

IAM - https://authn.vodafone.com.au, https://my.vodafone.com.au/createanaccount

- Auth0 Events Reference: https://auth0.com/docs/customize/log-streams/event-filters
- Auth0 Sample Events: https://tpgtelecom.sharepoint.com/:f:/r/sites/CMID/Shared%20Documents/General/Release%203/3.%20Segment%20CDP/2.%20Design/INPUT/Source%20-%20Auth0/auth0-sample-events?csf=1&web=1&e=vPEdXd
- Lambda Example: https://tpgtelecom.sharepoint.com/:t:/r/sites/CMID/Shared%20Documents/General/Release%203/3.%20Segment%20CDP/2.%20Design/INPUT/Source%20-%20Auth0/auth0-event-logs-aws-lambda.txt?csf=1&web=1&e=knGnQX

1. **User Logged In**

2.
```
{
  "type": "identify",
  "userId": "auth0|66c82e7b6231a2d8cb0f3468",
  "traits": {
    "email": "john@domain.com",
    "user_name": "john.smith",
    "identity_source": "Auth0 Login",
    "identity_confidence": "explicit",
    "auth_event": "user_logged_in",
    "brand": "vodafone",
    "environment": "prod"
  }
}
{
  "type": "track",
  "event": "Login Completed",
  "userId": "auth0|66c82e7b6231a2d8cb0f3468",
  "properties": {
    "authentication_provider": "Auth0",
    "login_method": "password_or_sso",
    "brand": "vodafone",
    "environment": "prod"
  }
```

1. **User Login Setup**

2.
```
{
  "type": "identify",
  "userId": "auth0|66c82e7b6231a2d8cb0f3468",
  "traits": {
    "email": "john@domain.com",
    "user_name": "john.smith",
    "identity_source": "Auth0 Login Setup",
    "identity_confidence": "explicit",
    "auth_event": "user_login_setup",
    "brand": "vodafone",
    "environment": "prod"
  }
}
{
  "type": "track",
  "event": "Login Setup Completed",
  "userId": "auth0|66c82e7b6231a2d8cb0f3468",
  "properties": {
    "identity_provider": "Auth0",
```

```
    "brand": "vodafone",
    "environment": "prod"
  }
```

1. **User Session Refreshed**

2.
```
{
  "type": "identify",
  "userId": "auth0|66c82e7b6231a2d8cb0f3468",
  "traits": {
    "identity_source": "Auth0 Session Refresh",
    "identity_confidence": "explicit",
    "auth_event": "session_refreshed",
    "brand": "vodafone",
    "environment": "prod"
  }
}
```

1. **Login Failed**

2.
```
{
  "type": "track",
  "event": "Login Failed",
  "properties": {
    "auth_provider": "auth0",
    "failure_reason": "invalid_credentials",
    "login_method": "password",
    "brand": "vodafone",
    "environment": "prod"
  }
}
```

1. **Login Setup Failed**

2.
```
{
  "type": "track",
  "event": "Login Setup Failed",
  "properties": {
    "auth_provider": "auth0",
    "failure_reason": "validation_error",
    "step": "username_creation",
    "brand": "vodafone",
    "environment": "prod"
  }
}
```

1. **Logout Completed**

```
{
  "type": "track",
  "event": "Logout Completed",
  "userId": "auth0|66c82e7b6231a2d8cb0f3468",
  "properties": {
    "auth_provider": "auth0",
    "logout_type": "user_initiated",
    "brand": "vodafone",
    "environment": "prod"
  }
}
```

1. **Logout Failed**

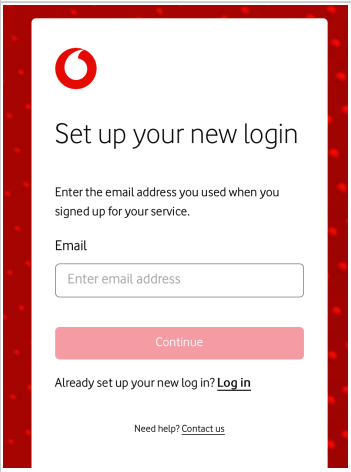
```
{
  "type": "track",
  "event": "Logout Failed",
  "userId": "auth0|66c82e7b6231a2d8cb0f3468",
  "properties": {
    "auth_provider": "auth0",
    "failure_reason": "session_not_found",
    "brand": "vodafone",
    "environment": "prod"
  }
}
Session Refresh Failed
```

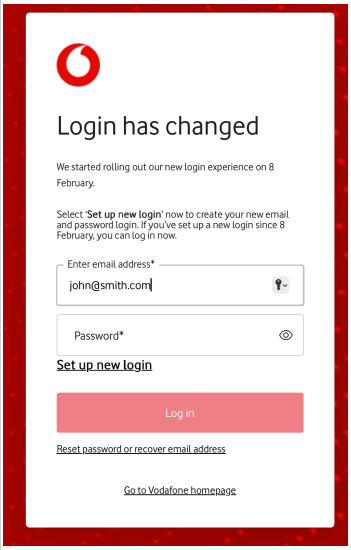
1. **Session Refresh Failed**

```
{
  "type": "track",
  "event": "Session Refresh Failed",
  "userId": "auth0|66c82e7b6231a2d8cb0f3468",
  "properties": {
    "auth_provider": "auth0",
    "failure_reason": "expired_token",
    "brand": "vodafone",
    "environment": "prod"
  }
}
```

Comments

Comment by Mushfiq Abdurrahman [02/May/26]





Comment by John Mathew [01/Jun/26]

Please align the requirement based on the events that Auth0 can pass and update the requirement excel with the right events.

https://tpgtelecom.sharepoint.com/:x:/r/sites/CMID/_layouts/15/Doc.aspx?sourcedoc=%7B80BA58E7-62DE-4241-9BDC-475947AA4072%7D&file=Segment_CDP_Environment_UAM_Design.xlsx&action=default&mobileredirect=true

R3.1 Digital to CDP Data → Data source CIAM

cc Ed Hor

FYI Poppy Wilson

Comment by Ani Paul [08/Sep/26]

- As per best practice, Segment Unify serves as the system of reference, while the actual user database or CRM remains the system of record. Since Segment ingests profile data from multiple sources, it can perform identity resolution and stitching based on the identifiers provided.
- Need clarification if the database of record can act as your ultimate `userId` mapping source else we might have more than 1 UserID and can lead to challenge with ID resolution (example store internal database primary key (UserId) directly inside the user's `app_metadata` in Auth0)
- Store Auth0 ID in a seperate identifier and perform transformation (shall be done in DWH/microservice) - confirmed

Comment by John Mathew [08/Sep/26]

Ani Paul Any autho or login event we can add additional attributes to be passed to Segment. As par of the Identify we will pass all the identity records which is made available to data warehouse from Siebel.

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